

Controlling Stepper Motors through Raspberry Pi



In the last issue of EFY we learnt how to control DC motors using Raspberry Pi, with the help of three projects. Let us now see how we can control stepper motors through Raspberry Pi, with the help of two projects.

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Robotics is never complete without stepper motors, and at one point or the other you have to run a stepper motor to do some complex job. In the last issue we discussed how to control DC motors through GPIOs of Raspberry Pi. In this article we discuss how to control stepper motors using Raspberry Pi, with two basic projects. In the first project we control a unipolar stepper motor using GPIOs of Raspberry Pi and in the second project we drive a bipolar stepper motor for finer control.

Project 1: Raspberry Pi controlling stepper motor

The GPIO pins can at most have 3.3 V at 60 mA, which is not enough to run a stepper motor. Therefore it is almost universal to use stepper motor along with a Darlington or H-bridge package, which acts as a switch to draw more power from additional source.

Stepper motors are mostly brushless motors having two coils. Most motors have five wires with the central connections of the two coils brought out as a single wire. Others have six wires where the central connections are kept apart. If the wires are not marked, with the help of a multimeter first identify both the coils.

Circuit and working

Fig. 2 shows the wiring diagram. The stepper motor used here is a 6-wire unipolar motor. Its central wires are joined together and connected to positive 5V supply (physical pin 2

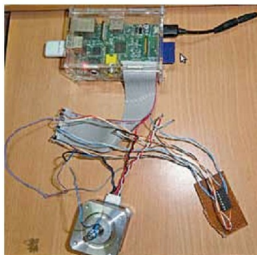


Fig. 1: Stepper motor control using Raspberry Pi

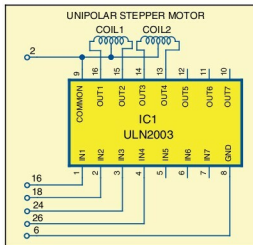


Fig. 2: Circuit for stepper motor control

of Raspberry Pi). External power for the stepper motor is applied at pins 8 and 9 of Darlington IC ULN2003 (IC1). Since the motor runs on 5 V, we just connected pin 9 of IC1 to pin 2 of Raspberry Pi.

Pins 16, 18, 24, 26 of Raspberry Pi

are the GPIO pins which will control our stepper motor. The speed of the motor is decided by the time delay we set in our program. If you see the program, you will see it is as simple as just shifting the GPIO pins from low to high, one after another. The simple sequence completes in four steps, but for a jerk-free smoother operation the eight-step sequence can also be considered.

Working of the circuit is simple. Once you run the program (explained under software sub-head), the stepper motor will immediately start rotating. You can easily modify the delay by changing the value of 'sx' in the program. If the stepper motor just vibrates without rotating, try to reverse one channel's output. If it still does not work, do the same for the other channel. To reverse the rotation, switch one wire of one channel with the other channel.

Software

The software for this project is designed in PHP. To run this program successfully, you need PHP and PHP-GPIO library.

First access the Raspberry Pi remotely so that you do not have to connect a separate monitor, keyboard and mouse to it. You can refer article 'Remote Access with Raspberry Pi' published in June issue to learn various ways of accessing it from your existing computer. Once you have accessed the